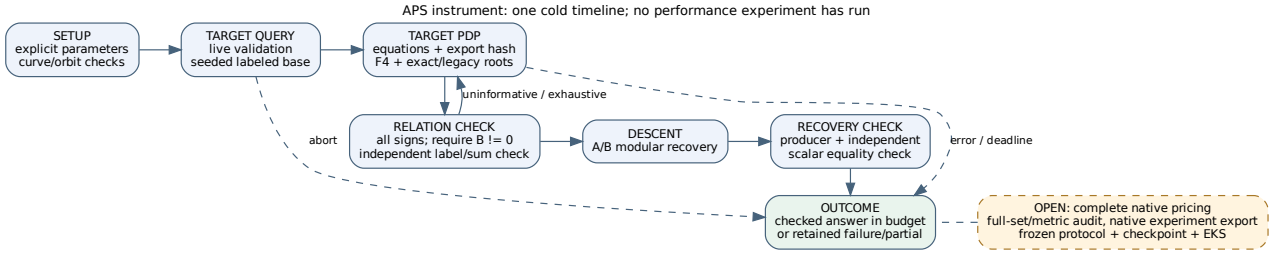
APS measurement instrument: implementation and open gates

Native ecbench continuation

30 October 2026

Outcome and scope. This adds a bounded native ecbench adapter for the four existing APS solver/encoding leads. It is instrumentation, not a measured improvement. No performance experiment has launched. Practical speed, scaling, complete operation cost and the full research panel remain unresolved.

What is charged. One monotonic timeline records target-independent setup, then all target-dependent work: validation and labeled-base sampling, polynomial construction and equation-export binding, F4 and root extraction, signed checks, modular recovery and independent affine verification. Setup plus the five exclusive online phases equals the cold solver interval exactly. All retries, pre-transcript aborts, failed branches and deadlines remain in the record. Final report aggregation/serialization is outside the algorithm interval; observer and equation-export construction overhead is inside and must be assessed.



Controls and correctness. Search parameters are explicit. The same constructor and seed support exhaustive/first-informative, legacy/exact roots, unordered cubic/line divisor and certified orbit line/pruning controls. Unsupported combinations are refused at planning. Every accepted witness is checked with separate ordinary-residue affine arithmetic: its labels, signed zero sum, nonzero denominator, modular equation and final scalar recovery. This attests a witness, not root completeness, execution history or costs.

Native-accounting limit. F4/root multiplication diagnostics and the independent check's group operations are retained. Core group/field work, polynomial construction, allocation and hashing are not completely counted or priced. GAE is consequently a lower bound; complete S and bitop/v1 rankings are inadmissible. The native-work-complete, cost-attested and independently complete-set-audited flags remain false. Closing a wall interval does not close this operation ledger.

Next evidence. Review a held-out four-size ladder within the existing prime-field bound, complete the native unified export and metric/full-output validation, and freeze the schema-valid protocol, controls, seeds, analysis, sample-size rationale and budget before affected data. EKS admission requires L2, A/A, verified source/image/binary identities and the applicable recovery contract. The mandatory source checkpoint is still blocked by the earlier explicit upload rejection; no alternate route or smaller upload evades it.

Sources and validation. The source-linked README, CHECKS and protocol draft accompany this report; final normalized receipts pin the code actually checked. Prior studies retain their original validation boundaries. This adds no theorem or Lean claim, paper-scale reproduction, new curve route or security conclusion. Development-test durations are not performance evidence.

Native adapter; existing APS solver and observers; APS manuscript/source map; SEC 1 v2.0, sections 2.2.1/3.2.2.1; ecbench standard.